Supplementary Materials - A Theoretical Perspective on Subnetwork Contributions to Adversarial Robustness

1st Joshua Andle
School of Computing and Information Science
University of Maine
Orono, United States
joshua.andle@maine.edu,

2nd Jovon Craig
School of Computing and Information Science
University of Maine
Orono, United States
jovon.craig@maine.edu

3rd Theodore S. Nowak
Coastal Sciences Division
Pacific Northwest National Laboratory
Sequim, United States
theodore.nowak@pnnl.gov,

4th Salimeh Yasaei Sekeh
Computer Science
San Diego State University
San Diego, United States
ssekeh@sdsu.edu,

I. Overview

The supplementary materials provided here are broken into two sections. First the complete proofs are provided for all theorems. In the second section we provide results of additional experiments which provide some supporting evidence to the results of the main paper. These experiments demonstrate the application of the theory in a linear setting. Additional results are also provided in the non-linear setting used for Algorithm 1 of the main paper under different adversarial attacks and network conditions to demonstrate consistency of results.

II. Proofs of Theorems

A. Proof of Theorem 1

First, we show the leftward implication, that if the layers $f^{(j)}, f^{(j-1)}, \ldots, f^{(1)}$ are semirobust, then $F^{(j)}$ is semirobust. This is proved because $F^{(j)}$ is $f^{(j)}(x^{(j-1)})$ where $x^{(j-1)} = f^{(j-1)} \circ \ldots \circ f^{(1)}$. Therefore, if $f^{(j)}$ is semirobust, regardless of whether any of $f^{(j-1)}, \ldots, f^{(1)}$ is semirobust, then $F^{(j)}$ is also semirobust.
Next, we show the rightward implication, that if $F^{(j)}$ is semirobust, then $f^{(j)}, f^{(j-1)}, \ldots, f^{(1)}$ are semirobust. If $F^{(j)}$ is semirobust then

$$\begin{aligned} &\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot G_j\circ F^{(j)}(\mathbf{X}+\delta)\right] \\ &= \mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot G_j\circ f^{(j)}(\mathbf{X}+\delta)\right] \geq \gamma_j \end{aligned} \tag{18}$$

This implies that $f^{(j)}$ is semirobust. Let $G_{j-1} = G_j \circ f^{(j)}$, then

$$\begin{aligned} &\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot G_j\circ F^{(j)}(\mathbf{X}+\delta)\right] \\ &= \mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot G_j\circ f^{(j)}\circ F^{(j-1)}(\mathbf{X}+\delta)\right] \\ &= \mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot G_{j-1}\circ F^{(j-1)}(\mathbf{X}+\delta)\right] \\ &= \mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot G_{j-1}\circ f^{(j-1)}(\mathbf{X}+\delta)\right] \geq \gamma_j, \end{aligned} \tag{19}$$

This implies that $f^{(j-1)}$ is semirobust. By induction, it's shown that the other layers $f^{(j-2)}, \ldots, f^{(1)}$ are also semirobust.

B. Proof of Lemma 1

Let $f^{(n-1)} = g \in \mathcal{L}_{n-1}$ and $f^{(n)} = h \in \mathcal{L}_n$. As $f^{(n-1)}$ is semirobust,

$$\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot G_{n-1}\circ g(\mathbf{X}+\delta)\right] \geq \gamma_{n-1}, \tag{20}$$

$$\text{and} \quad \sum_y \pi(y) I\left(g_\delta; h_\delta|y\right) \geq \rho, \tag{21}$$

and after simplification, we have

$$\begin{aligned} &\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot G_{n-1}\circ g(\mathbf{X}+\delta)\right] \\ &= \sum_y y\cdot\pi(y)\int \inf_{\delta\in S} D(x|y)\cdot G_{n-1}\circ g(x+\delta)d\mathbf{x} \end{aligned} \tag{22}$$

with $\pi(y)$ being the prior of y, and $D(x|y)$ being the probability density function of x and y. Let $g_\delta = g(\mathbf{x}+\delta) \in \mathcal{L}_{n-1}$, with components $g_\delta^{(i)} = g_i(\mathbf{x}+\delta)$ such that

$\mathbf{x} = (\mathbf{x^{(1)}}, \ldots, \mathbf{x^{(d)}}) \in \mathcal{R}^{\mathbf{d}}$, and $g = (g_1, \ldots, g_m) \in \mathcal{L}_{n-1}$. Note that the multivariate transformation g_i is one to one; hence, the transformation is invertible and can be solved for the equation $x^{(i)} + \delta^{(i)} = g_i^{-1}(g_\delta)$.

Thus, the last line in (22) equals

$$\sum_y y \cdot \pi(y) \int \inf_{\delta \in S} D(g^{-1}(g_\delta) - \delta|y) \cdot G_{n-1} \circ g_\delta \; |J| \; dg_\delta$$

where $|J|$ denotes the absolute value of the determinant of the Jacobian J. In addition, using the probability density for a function of a random variable, we can write:

$$\begin{aligned} &\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot G_{n-1} \circ g(\mathbf{X}+\delta)\right] \\ &= \sum_y y \cdot \pi(y) \int \inf_{\delta \in S} p(g_\delta|y) \cdot G_{n-1} \circ g_\delta \; dg_\delta \end{aligned} \tag{23}$$

By simplifying mutual information $I(g_\delta; h_\delta|y)$, using $\log(x) \leq x+1$, and recalling assumption **B1**:

$$\begin{aligned} &\sum_y \pi(y) \iint \inf_{\delta \in S} p(g_\delta, h_\delta|y) \\ &\qquad \left[\frac{p(g_\delta, h_\delta|y)}{p(g_\delta|y)p(h_\delta|y)}\right] \cdot dg_\delta dh_\delta + 1 \geq \rho \end{aligned} \tag{24}$$

To show that $f^{(n)}$ is γ_n-semirobust, we prove

$$\gamma_n \leq \mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot h(\mathbf{X}+\delta)\right].$$

Set $\gamma_n \leq \gamma_{n-1} + \rho$, hence we need to show that

$$\gamma_{n-1} + \rho \leq \sum_y y \cdot \pi(y) \int \inf_{\delta \in S} p(h_\delta|y) \cdot h_\delta dh_\delta \tag{25}$$

Following (24) and semirobustness for $f^{(n-1)}$, the inequality (25) can be transformed into

$$\begin{aligned} &\sum_y \pi(y) \iint \inf_{\delta \in S} p(g_\delta, h_\delta|y) \; \cdot \Big(y \cdot G_{n-1} \circ g_\delta \\ &\quad + \frac{p(g_\delta, h_\delta|y)}{p(g_\delta|y)p(h_\delta|y)} - y \cdot h_\delta + 1\Big) \cdot dg_\delta dh_\delta \leq 0 \end{aligned} \tag{26}$$

and subsequently into

$$\sum_y \pi(y) \mathbb{E}_{p(g_\delta, h_\delta|y)}[\inf_{\delta \in S} y \cdot (G_{n-1} \circ g_\delta - h_\delta)] \tag{27}$$

which holds true and is upper bounded by $\leq -(1+U)$, recalling the assumption **B2**. This concludes the proof of the lemma.

C. Proof of Theorem 2

To prove Theorem 2, given that

$$\begin{aligned} &\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot G_a \circ h_a(\mathbf{X}+\delta)\right] \geq \gamma_a \;\text{ and} \\ &\sum_y \pi(y) I\left(h_{\delta,a}; h_{\delta,a+1}|y\right) \geq \rho_{a+1}, \end{aligned} \tag{28}$$

From the inequalities above we show that for $\gamma_{a+1} \leq \gamma_a + \rho_{a+1}$

$$\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot G_{a+1} \circ h_{a+1}(\mathbf{X}+\delta)\right] \geq \gamma_{a+1} \tag{29}$$

Under assumptions **A1** and **A2** for $j = a+1$, we can simplify (28) and (29). We then need to show

$$\begin{aligned} &\sum_y \pi(y) \mathbb{E}_{p(g_\delta, h_{\delta,a+1}|y)}[\inf_{\delta \in S} y \cdot \\ &\qquad (G_a \circ h_{\delta,a} - G_{a+1} \circ h_{\delta,a+1}) \leq -(1+U_{a+1}) \end{aligned} \tag{30}$$

The above holds true recalling the assumption **A2** and $\pi(y)$ being non-negative. Hence, $f^{(a+1)}$ is γ_{a+1}-semirobust. And because $f_a = F^{(a)}$ is γ_a-semirobust, then according to Theorem 1, $F^{(a+1)}$ is also γ_{a+1}-semirobust.

Similarly, since $f_a = F^{(a+1)}$ is γ_{a+1}-semirobust, and by assumptions **A1** and **A2** for $j = a+2$, it is implied that $f^{(a+2)}$ is γ_{a+2}-semirobust. Recursively, it can be shown that all layers in f_b, i.e. $f^{(a+1)}, \ldots, f^{(n)}$, are γ_j-semirobust for $j = a+1, \ldots, n$ respectively. Then, according to Theorem 1, f_b is γ_b-semirobust where $\gamma_b \leq \gamma_a + \sum_{j=a+1}^{b} \rho_j$, proving Theorem 2.

D. Proof of Lemma 2

$$\begin{aligned} &\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot f^{(n)}(\mathbf{X}+\delta)\right] \\ &= \mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot \sum_{i=1}^{n-1} \lambda_i^T \cdot f^{(i)}(\mathbf{X}+\delta)\right] \end{aligned} \tag{31}$$

$$\begin{aligned} &\sum_{i=1}^{n-1} \mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot \lambda_i^T f^{(i)}(\mathbf{X}+\delta)\right] \\ &= \sum_{i=1}^{n-1} \mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot \lambda_i^T F^{(i)}(\mathbf{X}+\delta)\right]. \end{aligned} \tag{32}$$

The last equality holds true because the noises are added to the input $\mathbf{X}$ and since in feed-forward network, each layer is a function of the previous layer therefore $f^{(i)}(\mathbf{X}+\delta) = F^{(i)}(\mathbf{X}+\delta)$.

Next, by letting $G_i = \lambda_i^T$, then we have

$$\begin{aligned} &\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot f^{(n)}(\mathbf{X}+\delta)\right] \\ &= \sum_{i=1}^{n-1} \mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot G_i \circ F^{(i)}(\mathbf{X}+\delta)\right] \geq \sum_{i=1}^{n-1} \gamma_i := \gamma_n. \end{aligned} \tag{33}$$

The first inequality is true because of Theorem 1. This concludes that $f^{(n)}$ is γ_n-semirobust.

E. Proof of Theorem 3

In Theorem 3, note that if $f_b = f^{(n-1)}$ and $f_a = f_{n-1} = F^{(1,n-1)}$, then it turns to Lemma 2. We prove the theorem where $f_b = F^{(n-1,n)}$ and $f_a = F^{(1,n-2)}$, and the general case f_a and f_b can be shown similarly by extension. Let $G_b : \mathcal{L}_b \mapsto \mathcal{Y}$ be a function that maps layer f_b to the output y. Proof of the case where $f_b = F^{(n-1,n)}$ and $f_a = F^{(n-2)}$:

$$\begin{aligned} &\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot F^{(n-1,n)}(\mathbf{X}+\delta)\right] \\ &= \mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta \in S} y \cdot f^{(n)}(\mathbf{X}+\delta)\right] \end{aligned} \tag{34}$$

Given that $f^{(n)}$ is a linear combination of all the other layers, with λ_{in}^T mapping $f^{(i)}$ to y,

$$\begin{aligned}
&=\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\sum_{i=1}^{n-1}\lambda_{in}^T\cdot f^{(i)}(\mathbf{X}+\delta)\right]\\
&=\sum_{i=1}^{n-1}\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{in}^T\cdot f^{(i)}(\mathbf{X}+\delta)\right]\\
&=\sum_{i=1}^{n-2}\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{in}^T\cdot f^{(i)}(\mathbf{X}+\delta)\right]\\
&+\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{n-1(n)}^T\cdot f^{(n-1)}(\mathbf{X}+\delta)\right]
\end{aligned}\tag{35}$$

Let $G_i=\lambda_{in}^T$, and let α be the second term in (35), i.e.

$$\alpha:=\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{n-1(n)}^T\cdot f^{(n-1)}(\mathbf{X}+\delta)\right],$$

therefore (35) equals

$$\sum_{i=1}^{n-2}\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot G_i\circ f^{(i)}(\mathbf{X}+\delta)\right]+\alpha.\tag{36}$$

Then, using Theorem 1, we have

$$\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{in}^T\cdot f^{(i)}(\mathbf{X}+\delta)\right]\geq\gamma_i,$$

which bounds (36) as

$$\begin{aligned}
&\sum_{i=1}^{n-2}\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot G_i\circ f^{(i)}(\mathbf{X}+\delta)\right]+\alpha\\
&\qquad\geq\sum_{i=1}^{n-2}\gamma_i+\alpha=\gamma_a+\alpha
\end{aligned}\tag{37}$$

where $\gamma_a=\sum_{i=1}^{n-2}\gamma_i$. Now simplifying α, given that $f^{(n-1)}$ is a linear combination of the layers before it, with $\lambda_{i(n-1)}^T$ mapping $f^{(i)}$ to $f^{(n-1)}$:

$$\mathbb{E}_{(\mathbf{X},y)\sim D}[\inf_{\delta\in S} y\cdot\lambda_{n-1(n)}^T\cdot f^{(n-1)}(\mathbf{X}+\delta)]$$

$$=\mathbb{E}_{(\mathbf{X},y)\sim D}[\inf_{\delta\in S} y\cdot\lambda_{n-1(n)}^T\cdot\sum_{i=1}^{n-2}\lambda_{i(n-1)}^T\cdot f^{(i)}(\mathbf{X}+\delta)]$$

$$=\sum_{i=1}^{n-2}\mathbb{E}_{(\mathbf{X},y)\sim D}[\inf_{\delta\in S} y]\cdot\lambda_{n-1(n)}^T\cdot\lambda_{i(n-1)}^T\cdot f^{(i)}(\mathbf{X}+\delta)].$$

Let $\tilde{G}_i=\lambda_{n-1(n)}^T\cdot\lambda_{i(n-1)}^T$. Then, using Theorem 1,

$$=\sum_{i=1}^{n-2}\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\tilde{G}_i\circ f^{(i)}(\mathbf{X}+\delta)\right]\geq\sum_{i=1}^{n-2}\gamma_i=\gamma_a$$

With both terms simplified, $\gamma_a+\alpha\geq\gamma_a+\gamma_a=\gamma_b$. Therefore, f_b is semirobust. This proof can be extended to any other combination of f_a and f_b. Let's show the case where $f_b=F^{(a+1,n)}$ and $f_a=F^{(a)}$:

$$\begin{aligned}
&\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot F^{(a+1,n)}(\mathbf{X}+\delta)\right]\\
&\qquad=\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot f^{(n)}(\mathbf{X}+\delta)\right]
\end{aligned}\tag{38}$$

Given that $f^{(n)}$ is a linear combination of all the other layers, with λ_{in}^T mapping $f^{(i)}$ to y,

$$\begin{aligned}
&=\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\sum_{i=1}^{n-1}\lambda_{in}^T\cdot f^{(i)}(\mathbf{X}+\delta)\right]\\
&=\sum_{i=1}^{n-1}\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{in}^T\cdot f^{(i)}(\mathbf{X}+\delta)\right]
\end{aligned}\tag{39}$$

We break down $\sum_{i=1}^{n-1}$ as $\sum_{i=1}^{a}+\sum_{i=a+1}^{n-1}$, therefore (39) equals

$$\begin{aligned}
&\sum_{i=1}^{a}\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{in}^T\cdot f^{(i)}(\mathbf{X}+\delta)\right]\\
&+\sum_{i=a+1}^{n-1}\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{in}^T\cdot f^{(i)}(\mathbf{X}+\delta)\right]
\end{aligned}\tag{40}$$

Let $G_i=\lambda_{in}^T$, and for $i=a+1,\ldots,n-1$ let α_i be

$$\alpha_i:=\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{in}^T\cdot f^{(i)}(\mathbf{X}+\delta)\right].\tag{41}$$

Then, by using Theorem 1 again we have,

$$\begin{aligned}
&\sum_{i=1}^{a}\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot G_i\circ f^{(i)}(\mathbf{X}+\delta)\right]\\
&+\sum_{i=a+1}^{n-1}\alpha_i\geq\sum_{i=1}^{a}\gamma_i+\sum_{i=a+1}^{n-1}\alpha_i=\gamma_a+\sum_{i=a+1}^{n-1}\alpha_i
\end{aligned}\tag{42}$$

where $\gamma_a=\sum_{i=1}^{a}\gamma_i$. Next we show that $\sum_{i=a+1}^{n-1}\alpha_i\geq\gamma_a\Big((n-1-a)(n-a)/2\Big)$, and conclude the proof by setting $\gamma_b:=\gamma_a+\gamma_a\Big((n-1-a)(n-a)/2\Big)$. Now by the assumption that

$$f^{(i)}=\sum_{\ell=1}^{i-1}\lambda_{\ell i}^T\cdot f^{(\ell)},\tag{43}$$

with $\lambda_{\ell i}^T$ mapping $f^{(\ell)}$ to $f^{(i)}$, then simplifying α_i yields

$$\begin{aligned}
\alpha_i&=\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{in}^T\sum_{\ell=1}^{i-1}\lambda_{\ell i}^T\cdot f^{(\ell)}(\mathbf{X}+\delta)\right]\\
&=\sum_{\ell=1}^{i-1}\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{in}^T\lambda_{\ell i}^T\cdot f^{(\ell)}(\mathbf{X}+\delta)\right]
\end{aligned}\tag{44}$$

for $i=a+1,\ldots,n-1$. Hence we have $\sum_{i=a+1}^{n-1}\alpha_i$ equals to

$$\sum_{i=a+1}^{n-1}\sum_{\ell=1}^{i-1}\mathbb{E}_{(\mathbf{X},y)\sim D}\left[\inf_{\delta\in S} y\cdot\lambda_{in}^T\cdot\lambda_{\ell i}^T\cdot f^{(\ell)}(\mathbf{X}+\delta)\right]$$

$$= \sum_{i=a+1}^{n-1} \sum_{\ell=1}^{i-1} \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{\ell n} \circ f^{(\ell)}(\mathbf{X}+\delta) \right] \quad (45)$$

where $G_{\ell n} := \lambda_{in}^T \cdot \lambda_{\ell i}^T$. Under the assumption (43), we know that for $i = a+1, \ldots, n-1$,

$$\sum_{\ell=1}^{a} \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{\ell n} \circ f^{(\ell)}(\mathbf{X}+\delta) \right] \geq \gamma_a, \quad (46)$$

$$\begin{aligned} \text{Therefore} \quad & \sum_{i=a+1}^{n-1} \alpha_i \geq (n-1-a)\gamma_a \\ & + \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{(a+1)n} \circ f^{(a+1)}(\mathbf{X}+\delta) \right] \\ & + \sum_{\ell=a+1}^{a+2} \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{\ell n} \circ f^{(\ell)}(\mathbf{X}+\delta) \right] \\ & + \ldots \\ & + \sum_{\ell=a+1}^{n-2} \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{\ell n} \circ f^{(\ell)}(\mathbf{X}+\delta) \right] \end{aligned} \quad (47)$$

Without loss of generality, assume that $G_{\ell n} = G_\ell$ for all $\ell = a+1, \ldots, n-2$. This simplifies (47) as

$$\begin{aligned} & \sum_{i=a+1}^{n-1} \alpha_i \geq (n-1-a)\gamma_a \\ & + \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{(a+1)} \circ f^{(a+1)}(\mathbf{X}+\delta) \right] \\ & + \sum_{\ell=a+1}^{a+2} \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{\ell} \circ f^{(\ell)}(\mathbf{X}+\delta) \right] \\ & + \ldots \\ & + \sum_{\ell=a+1}^{n-2} \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{\ell} \circ f^{(\ell)}(\mathbf{X}+\delta) \right] \end{aligned} \quad (48)$$

Below we show that if $f_a = F^{(1,a)}$ is semirobust and $f^{(a+1)}$ is a linear combination of layers $f^{(1)}, \ldots, f^{(a)}$, with $\lambda_{\ell(a+1)}$ mapping $f^{(\ell)}$ to $f^{(a+1)}$, then $f_{a+1} = F^{(1,a+1)}$ is a semirobust feature:

$$\begin{aligned} & \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{(a+1)} \circ f^{(a+1)}(\mathbf{X}+\delta) \right] \\ & = \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{(a+1)} \circ \sum_{\ell=1}^{a} \lambda_{\ell(a+1)}^T f^{(\ell)}(\mathbf{X}+\delta) \right] \\ & = \sum_{\ell=1}^{a} \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{(a+1)} \circ \lambda_{\ell(a+1)}^T f^{(\ell)}(\mathbf{X}+\delta) \right] \\ & = \sum_{\ell=1}^{a} \mathbb{E}_{(\mathbf{X},y)\sim D} \left[\inf_{\delta \in S} y \cdot G_{\ell} \circ f^{(\ell)}(\mathbf{X}+\delta) \right] \end{aligned} \quad (49)$$

where $G_\ell = G_{(a+1)} \circ \lambda_{\ell(a+1)}^T$. Since f_a is semirobust, all layers $f^{(1)}, \ldots, f^{(a)}$ are semirobust. Hence, the right-hand side in (49) is greater than or equal to $\sum_{\ell=1}^{a} \gamma_\ell = \gamma_a$.
Consequently, with the same methodology, this can be extended to the following: if $f_a = F^{(1,a)}$ is semirobust, and $f^{(a+\ell)}$ is a linear combination of layers $f^{(1)}, \ldots f^{(a+\ell-1)}$, then $f_{a+\ell} = F^{(1,a+\ell)}$ for $\ell = 1, \ldots, n-2-a$ is semirobust. This implies that the Ineq. (47) is lower-bounded by

$$\begin{aligned} & (n-1-a)\gamma_a + \gamma_a + \underbrace{\sum_{\ell=a+1}^{a+2} \gamma_a}_{2\times\gamma_a} \\ & + \underbrace{\sum_{\ell=a+1}^{a+3} \gamma_a}_{3\times\gamma_a} + \ldots + \underbrace{\sum_{\ell=a+1}^{n-2} \gamma_a}_{(n-2-a)\times\gamma_a}, \end{aligned}$$

$$\begin{aligned} \text{which is equal to} \quad & (n-1-a)\gamma_a + \gamma_a \sum_{j=1}^{n-2-a} j \\ & = (n-1-a)\gamma_a + \gamma_a \Big((n-2-a)(n-1-a) \big/ 2 \Big). \end{aligned} \quad (50)$$

This proves that $\sum_{i=a+1}^{n-1} \alpha_i \geq \gamma_a \Big((n-1-a)(n-a)/2 \Big) = \gamma_b$.

III. Supplementary Experiments

A. Training Details

Training for CNNs (both regular and adversarial) is done with a Stochastic Gradient Descent optimizer for 120 epochs with a learning rate of 0.01 which decreases by a factor of 10 at epochs 40 and 80, using a weight decay of 5×10^{-4}. Training for transformers is done using the AdamW optimizer for 120 epochs with weight decay of 0.0005, and learning rate decay scheduled by cosine annealing, beginning at 0.0001. We use a batch size of 512 for CIFAR-10 and CIFAR-100, and 256 for Tiny Imagenet. For adversarial training we use PGD with 10 iterations, $\epsilon = \frac{8}{255}$, and step size of $\frac{2}{255}$ under an L_∞-norm bound.

B. Linear Implementation

In order to implement and demonstrate the concept in Theorem 3 regarding a linear setting for semirobustness, we have outlined an algorithm that predicts the labels of each input as a linear combination of the activation values of the layers in the semirobust subnetwork f_a^*. Algorithm S1 details how we apply this linearity constraint on the dependency between layers in the subnetworks. For a full network (f_a^*, f_b^*), rather than generating predictions as the outputs of the nonlinear subnetwork f_b^*, we instead seek to find a set of parameters λ such that we can take a linear combination of the activations within f_a^* to directly predict the labels. By calculating the optimal λ values we aim to achieve the baseline robust test accuracy Acc^* on the adversarial data using the frozen f_a^* subnetwork. We outline how we've addressed the problem of finding these optimal λ values through a straightforward linear algebra approach to directly solve for them.

Given a network which has been adversarially trained to be fully robust, denoted as (f_a^*, f_b^*), the approach is to record the outputs from all layers in the frozen f_a^* subnetwork when applied to test data as $\mathbb{F}_a$. We also record the outputs of the

final layer in f_b^* which indicate the predicted labels of the fully robust network as $\mathbb{F}^{(n)}$. We then utilize the loss function

$$\ell = \sum_{j=a+1}^{n} ||f^{(j)*} - f^{(j)}|| \tag{51}$$

to set $\mathbb{F}^{(j)} = \mathbb{F}^{(j)*}$ for $j = n$ and solve for the λ values which minimizes the loss in the following steps:

$$\begin{aligned} \mathbb{F}_a^* \cdot \lambda_j &= \mathbb{F}^{(j)} \\ \mathbb{F}_a^* \cdot \lambda_j &= \mathbb{F}^{(j)*} \\ {\mathbb{F}_a^*}^{-1} \cdot \mathbb{F}_a^* \cdot \lambda_j &= {\mathbb{F}_a^*}^{-1} \cdot \mathbb{F}^{(j)} \\ \mathbb{I} \cdot \lambda_j = \lambda_j &= {\mathbb{F}_a^*}^{-1} \cdot \mathbb{F}^{(j)} \end{aligned} \tag{52}$$

Algorithm S1 Determining Hyperparameter λ

Do subsequent regular and adversarial training of $F^{(n)}$ as (f_a, f_b) and (f_a^*, f_b^*) respectively
Replace f_b^* with the linear-combination $f^{(n)} = \sum_{i=1}^{a} \lambda_i^T \cdot f^{(i)}$
Initialize list λ of batch-wise calculations of λ_k
Initialize list $\mathbb{F}_a^*$ of batch-wise activation values $\mathbb{F}_{a,k}^*$
Initialize list $\mathbb{F}^{(j)}$ of batch-wise outputs of (f_a^*, f_b^*)
for each batch = k of testing data **do**
  Get outputs of (f_a^*, f_b^*) as $\mathbb{F}_k^{(j)}$ and append to $\mathbb{F}^{(j)}$
  Calculate the (f_a^*) activations $\mathbb{F}_{a,k}^*$ and append to $\mathbb{F}_a^*$
  Solve for $\lambda_k = {\mathbb{F}_{a,k}^*}^{-1} \cdot \mathbb{F}_k^{(j)}$
  Append λ_k to list λ
end
Take element-wise average values of λ_k across each index of λ to get the optimal values λ^*
Calculate test accuracy $\widetilde{Acc}$ of outputs $f^{(n)} = \sum_{i=1}^{a} {\lambda_i^*}^T \cdot f^{(i)}$
Report $\widetilde{Acc}$

In this way we calculate the optimal of λ through a matrix multiplication between the inverse of the activations in subnetwork f_a^* and the outputs from the fully robust network's final layer. In order to make this approach more tractable we split the problem of solving for λ over several batches of data. For each batch, we calculate the activations $\mathbb{F}_{a,k}^*$ and $\mathbb{F}_k^{(j)}$ for the current batch of data before using them to solve for a set of optimal λ_k for the given batch. After all batches are finished the average λ values are found across all batches to get λ^*. The baseline test accuracy Acc^* obtained with the nonlinear, fully robust model (f_a^*, f_b^*) is compared against that obtained by applying $f^{(n)} = \sum_{i=1}^{a} {\lambda_i^*}^T \cdot f^{(i)}$, denoted as $\widetilde{Acc}$. The goal of this algorithm is not to train a linear network capable of generalizing to new data, but to demonstrate that for a given semirobust subnetwork f_a^*, a set of λ can be found which approach robust accuracy when making predictions on adversarially attacked data, indicating full-network robustness.

Table S1 shows the results of applying Algorithm S1 to ResNet-18 across multiple settings. The reported value of "# f_a" indicates the size of f_a^* which is used for making predictions through the linear combination with λ. For ResNet-18, which has 18 convolutional and linear layers, if the size is 17 then it indicates that only the final linear layer is being replaced. This is the setting which most closely follows Theorem 3, however we also compare the accuracies obtained when using fewer layers in f_a^* to directly predict the label predictions in layer $f^{(n)}$. We report the accuracy Acc^* of the network (f_a^*, f_b^*), which is the regularly calculated test accuracy of the fully robust network. We then report the accuracy when predictions are calculated instead as $\sum_{i=1}^{a} {\lambda_i^*}^T \cdot f^{(i)}$. We compare the resulting accuracy against the robust baseline and a negative control calculated using random λ values.

Attack	# f_a^*	Acc^* (%)	$\widetilde{Acc}$ (%)	Diff. (%)	Acc_{rand}
None	17	69.7	61.59	-8.11	9.93
	13	69.7	60.64	-9.06	10.17
	9	69.7	58.47	-11.23	8.58
FGSM	17	43.08	42.91	-0.17	14.67
	13	43.08	42.46	-0.62	12.77
	9	43.08	41.19	-1.89	8.07
PGD	17	38.63	40.41	1.78	8.91
	13	38.63	40.13	1.5	10.1
	9	38.63	39.47	0.84	6.82
AA	17	34.11	48.5	14.39	7.12
	13	34.11	48.29	14.18	9.63
	9	34.11	47.15	13.04	9.75

TABLE S1: **Results of Algorithm S1 on CIFAR-10 with ResNet-18:** This table shows the resulting test accuracies of ResNet-18 on CIFAR-10 under varying attacks. We apply Algorithm S1 across three sizes of f_a^* for each attack, indicating how many layers are utilized when implementing λ. The test accuracy of (f_a^*, f_b^*) is reported as Acc^* while the accuracy of $\sum_{i=1}^{a} {\lambda_i^*}^T \cdot f^{(i)}$ is reported as $\widetilde{Acc}$. The accuracy when using randomly initialized λ values is reported for further comparison.

On weaker attacks predictions made under this linear setting are less accurate than the robust network. This difference decreases when adversarial attacks are applied, and for stronger attacks the predictions made with the linear setting outperform the baseline. For all settings as fewer layers are included in f_a^* the performance of the linear predictions decreases.

There are intuitive explanations for these observations. In this setting the aim is to demonstrate that the semirobust network f_a^* can be used to gain robustness under a constraint of linearity. We don't make any assumption about the structure of f_b, such as only having access to the previous layer, and as such we include all convolutional and linear layers in f_a^* when calculating $\sum_{i=1}^{a} {\lambda_i^*}^T \cdot f^{(i)}$. It's expected that replacing the non-linear portion of the network f_b with the linear combination using λ would reduce accuracy, which we see for clean data, however the results suggest that for more difficult attacks having access to the outputs of all layers in the subnetwork f_a^* outweighs

this detriment. These observations are further supported by the decrease in accuracy as the size of f_a^* decreases, in which case fewer layers' information is available when taking a linear combination resulting in worse performance.

These observations support the concepts in Theorem 3 that the semirobust subnetwork f_a^* can be used under a linear constraint to achieve full-network robustness. This experiment serves primarily to demonstrate the theory and has practical limitations. Although directly solving for λ in this way highlights that such a λ exists, the calculation involved would become intractable on larger datasets where you need to take the inverse of a matrix of all intermediate activations of the network. Additionally, we rely on the outputs of the non-linear layers f_b^* when solving for λ^* which requires already having a trained, fully robust network.

C. Additional Experiments

Algorithm 1 on Varying Attacks As a supplement to the results from the experiments on AutoAttack, we provide here additional results obtained across clean data, FGSM, and PGD. We report the results of Algorithm 1 in Tables S2-S4. These results span the use of VGG16, ResNet-18, and WideResNet-34-10 on CIFAR-10, CIFAR-100, and Tiny Imagenet. We ran experiments with either an f_b size of 4 or 12 layers, indicating the number of convolutional or linear layers included in f_b. We report the non-robust accuracy Acc, the baseline robust accuracy Acc^*, the initial semirobust accuracy of (f_a^*, f_b) as Acc_{sr}, and the best resulting accuracy of $(f_a^*, \widetilde{f_b})$ as $\widetilde{Acc}$. We additionally report the difference in accuracy $(\widetilde{Acc} - Acc^*)$, the average number of epochs required to converge, and a subset of the ρ values. If no trials converged for a given setting, the largest ρ values are reported over all trials as an approximation of the thresholds for a fully robust network $(f_a^*, \widetilde{f_b})$ following the observations that adversarial training increases ρ values as accuracy improves.

The results in Tables S2-S4 reflect several of the observations already made for AutoAttack. Larger semirobust subnetwork f_a^* sizes consistently yielded higher values of $\widetilde{Acc}$, which empirically suggests the intuitive case that when f_a^* constitutes an insufficient portion of the overall network, it may be less capable of conferring robustness to the full network. We do however see cases in PGD-attacked data where smaller f_a^* layers resulted in higher values of $\widetilde{Acc}$. Notably, when considering the ability of (f_a^*, f_b) to reach Acc^* through finetuning, compared to the results reported for AutoAttack far fewer settings on FGSM or PGD managed to match or outperform the baseline Acc^*. This may be because (f_a^*, f_b^*) was adversarially trained with PGD, leading to a higher level of robustness against it and similar attacks which is harder for the semirobust network (f_a^*, f_b) to reach in the small number of epochs used for each trial. Regarding the reported values of ρ, we see the same patterns which we have shown in the previous experiments, including similar ρ values across different attacks, smaller sizes of f_b yielding higher ρ values for a given layer, and lower values being reported for CIFAR-100 and ResNet-18 relative to CIFAR-10 and WideResNet-34-10.

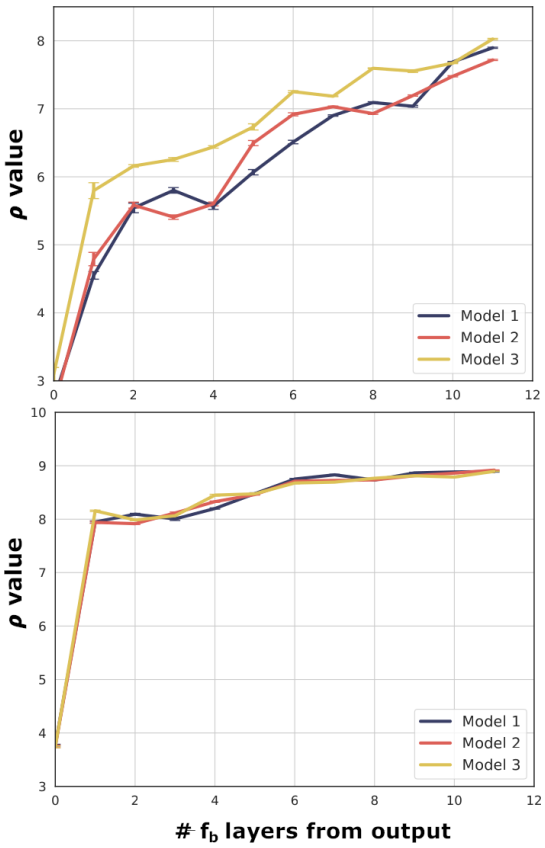

Fig. S1: **Triplicate experiment ρ values:** For ResNet-18 (left) and WideResNet-34-10 (right) on CIFAR-10, the mean ρ values are plotted for the trials of Algorithm 1 across three separately pretrained models.

Stability of ρ Across Trials: As Algorithm 1 performs trials over the same initial pretrained model, the results for it are reported among all trials for a given pretrained network. To ensure that this didn't lead to erroneous results we ran additional experiments comparing the ρ values obtained for trials run across multiple pretrained models. The results of these experiments for ResNet-18 and WideResNet-34-10 on FGSM-attacked CIFAR-10 are reported in Figure S1. For each setting, triplicate models were initialized and pretrained to yield different (f_a, f_b) and (f_a^*, f_b^*). Trials were then run for each pretrained network, and the average ρ values among trials were plotted for each pretrained network. We observe that when the trials are run on different sets of pretrained weights, resulting ρ values are generally similar. It's necessary to note that significant differences were observed among duplicates of ResNet-18 in Figure S1, however the previously reported trends in ρ values are still seen and the difference between duplicates is less than that between networks or datasets as demonstrated in Tables S2-S4 and Figure S1.

Impact of Number of Layers in f_b: Figure S2 demonstrates

Net	Dataset	# f_b	Acc	Acc^*	Acc_{sr}	$\widetilde{Acc}$	Diff.	Epochs	ρ_n	ρ_{n-3}	ρ_{n-7}	ρ_{n-11}
VGG16	CF-10	4	88.9	77.9	69.5	77.8	-0.1	-	2.3	6.7	-	-
		12	88.9	77.9	54.8	72.4	-5.5	-	2.2	6.3	8.0	8.5
	CF-100	4	61.9	48.7	39.1	48.9	-0.2	-	3.5	4.7	-	-
		12	61.9	48.7	21.8	45.1	-3.6	-	3.4	4.6	5.2	5.4
	TI	4	37.4	27.8	4.3	26.8	-0.9	-	3.2	4.0	-	-
		12	37.4	27.8	5.7	**29.8**	**2.0**	**2.1**	3.0	4.0	4.3	4.4
RN-18	CF-10	4	86.4	69.7	67.7	**71.4**	**1.8**	**2.0**	3.4	6.3	-	-
		12	86.4	69.7	62.5	66.3	-3.4	-	2.8	5.9	6.9	7.9
	CF-100	4	57.4	46.8	39.5	46.4	-0.3	-	3.6	4.6	-	-
		12	57.4	46.8	31.4	42.2	-4.6	-	3.6	4.6	4.8	5.0
	TI	4	33.4	29.7	8.0	28.6	-1.0	-	3.3	4.00	-	-
		12	33.4	29.7	3.2	**30.5**	**0.8**	**3.3**	3.3	4.0	4.1	4.2
WRN-34	CF-10	4	90.9	81.3	74.2	**81.6**	**0.3**	**1.0**	4.2	8.8	-	-
		12	90.9	81.3	68.9	77.7	-3.6	-	3.8	8.1	8.8	8.9
	CF-100	4	69.0	56.2	46.4	55.9	-0.4	-	3.9	5.6	-	-
		12	69.0	56.2	40.0	52.0	-4.2	-	3.7	5.5	5.6	5.6
	TI	4	47.8	45.2	7.2	43.1	-2.1	-	3.7	5.4	-	-
		12	47.8	45.2	3.2	38.9	-6.3	-	3.5	5.3	5.5	5.7

TABLE S2: **Algorithm 1 results evaluated for Non-Adversarial Data:** The summarized results of running Algorithm 1 for all CNN architectures and datasets are shown for clean (non-attacked) data. Accuracy on clean data for adversarially trained networks was used to determine the values of Acc^*, Acc_{sr}, and $\widetilde{Acc}$. Non-robust accuracy Acc is additionally reported for reference. The difference between $\widetilde{Acc}$ and Acc^*, as well as the average number of epochs required to reach the baseline robust accuracy are recorded. If no trials converged within 10 epochs, this value is omitted. The lowest ρ values for a subset of the layers in $\widetilde{f}_b$ are reported from amongst the trials which converged with Acc^*, where ρ_n corresponds to the last pair of layers in $\widetilde{f}_b$.

Net	Dataset	# f_b	Acc	Acc^*	Acc_{sr}	$\widetilde{Acc}$	Diff.	# Epochs	ρ_n	ρ_{n-3}	ρ_{n-7}	ρ_{n-11}
VGG16	CF-10	4	11.9	44.9	33.2	**45.0**	**0.1**	**7.8**	2.4	6.6	-	-
		12	11.9	44.9	17.7	43.1	-1.7	-	2.5	6.3	7.9	8.6
	CF-100	4	5.4	19.0	14.6	**19.0**	**0.0**	**8.8**	3.5	4.7	-	-
		12	5.4	19.0	4.9	**19.6**	**0.6**	**4.7**	3.4	4.5	5.2	5.4
	TI	4	0.3	5.6	1.0	5.3	-0.3	-	3.2	4.0	-	-
		12	0.3	5.6	0.7	**9.4**	**3.8**	**1**	3.0	4.0	4.3	4.4
RN-18	CF-10	4	6.2	43.1	34.4	41.3	-1.7	-	3.3	6.3	-	-
		12	6.2	43.1	22.5	38.1	-4.9	-	2.9	5.9	6.9	7.9
	CF-100	4	4.4	19.8	14.9	19.2	-0.6	-	3.6	4.7	-	-
		12	4.4	19.8	7.6	18.5	-1.3	-	3.6	4.7	4.8	5.0
	TI	4	0.1	6.6	1.5	**6.7**	**0.1**	**4**	3.3	4.0	-	-
		12	0.1	6.6	0.7	**11.1**	**4.5**	**1**	3.3	4.0	4.1	4.2
WRN-34	CF-10	4	4.1	50.1	39.8	49.8	-0.3	-	4.0	8.8	-	-
		12	4.1	50.1	34.9	45.8	-4.3	-	3.8	8.0	8.8	8.9
	CF-100	4	3.0	24.5	19.3	23.9	-0.5	-	4.0	5.6	-	-
		12	3.0	24.5	13.8	23.4	-1.0	-	3.7	5.5	5.6	5.6
	TI	4	0.1	19.3	1.8	19.1	-0.2	-	3.9	4.8	-	-
		12	0.1	19.3	1.0	17.8	-1.5	-	3.6	4.3	4.7	4.8

TABLE S3: **Algorithm 1 results evaluated for FGSM-Attacked Data:** The summarized results of running Algorithm 1 for all CNN architectures and datasets are shown for test data attacked with FGSM. Accuracy on this data was used to determine the values of Acc^*, Acc_{sr}, and $\widetilde{Acc}$. Non-robust accuracy Acc is additionally reported for reference. The difference between $\widetilde{Acc}$ and Acc^*, as well as the average number of epochs required to reach the baseline robust accuracy are recorded. If no trials converged within 10 epochs, this value is omitted. The lowest ρ values for a subset of the layers in $\widetilde{f}_b$ are reported from amongst the trials which converged with Acc^*, where ρ_n corresponds to the last pair of layers in $\widetilde{f}_b$.

the effect of selecting different sizes of f_b during Algorithm 1. When f_b comprised of 4 layers slightly higher ρ values were observed in most experiments, while the difference between 8 layers and 12 were minor. For this reason we focus on comparing conditions at 4 and 12 layers for the size of f_b in CNNs.

Impact of Stopping Condition k on ρ: We run experiments across varying values of k in Algorithm 1 to test its impact on ρ. The results reported in Figure S3 demonstrate that the values of ρ increase as the stopping criteria is made more strict. This demonstrates a similar trend as is seen when plotting MI across the epochs of a given trial, as one would expect a looser stopping criteria to result in stopping at an earlier finetuning epoch.

Net	Dataset	# f_b	Acc	Acc^*	Acc_{sr}	$\widetilde{Acc}$	Diff.	Epochs	ρ_n	ρ_{n-3}	ρ_{n-7}	ρ_{n-11}
VGG16	CF-10	4	0.2	44.6	33.0	44.5	-0.1	-	2.5	6.7	-	-
		12	0.2	44.6	16.9	44.2	-0.4	-	2.5	6.3	7.9	8.5
	CF-100	4	0.2	18.2	14.3	18.0	-0.2	-	3.5	4.7	-	-
		12	0.2	18.2	4.5	**19.6**	**1.4**	**2.5**	3.4	4.6	5.2	5.4
	TI	4	0	5.0	0.9	4.9	-0.1	-	3.2	4.0	-	-
		12	0	5.0	0.5	**10.4**	**5.4**	**1**	3.0	4.0	4.3	4.4
RN-18	CF-10	4	0	38.5	29.8	36.5	-2.0	-	3.3	6.3	-	-
		12	0	38.5	16.9	33.7	-4.8	-	2.9	5.9	6.9	7.9
	CF-100	4	0	16.0	12.3	15.9	-0.1	-	3.6	4.6	-	-
		12	0	16.0	5.5	**16.3**	**0.3**	**9.2**	3.6	4.7	4.8	5.1
	TI	4	0	6.1	1.6	**6.4**	**0.4**	**1**	3.3	4.0	-	-
		12	0	6.1	0.7	**12.1**	**6.0**	**1**	3.3	4.0	4.1	4.2
WRN-34	CF-10	4	0	43.2	34.5	42.8	-0.4	-	4.0	8.8	-	-
		12	0	43.2	30.3	41.4	-1.8	-	3.8	8.1	8.9	8.9
	CF-100	4	0	20.4	17.0	19.6	-0.8	-	3.9	5.7	-	-
		12	0	20.4	11.1	**20.5**	**0.1**	**6.0**	3.7	5.4	5.6	5.6
	TI	4	0	13.1	1.8	13.0	-0.1	-	3.6	4.7	-	-
		12	0	13.1	1.1	**16.0**	**3.0**	**1**	3.5	4.5	4.6	4.6

TABLE S4: **Algorithm 1 results evaluated for PGD-Attacked Data:** The summarized results of running Algorithm 1 for all CNN architectures and datasets are shown for data attackd with PGD. Accuracy of adversarially trained networks on PGD-attacked test data was used to determine the values of Acc^*, Acc_{sr}, and $\widetilde{Acc}$. Non-robust accuracy Acc is additionally reported for reference. The difference between $\widetilde{Acc}$ and Acc^*, as well as the average number of epochs required to reach the baseline robust accuracy are recorded. If no trials converged within 10 epochs, this value is omitted. The lowest ρ values for a subset of the layers in $\widetilde{f_b}$ are reported from amongst the trials which converged with Acc^*, where ρ_n corresponds to the last pair of layers in $\widetilde{f_b}$.

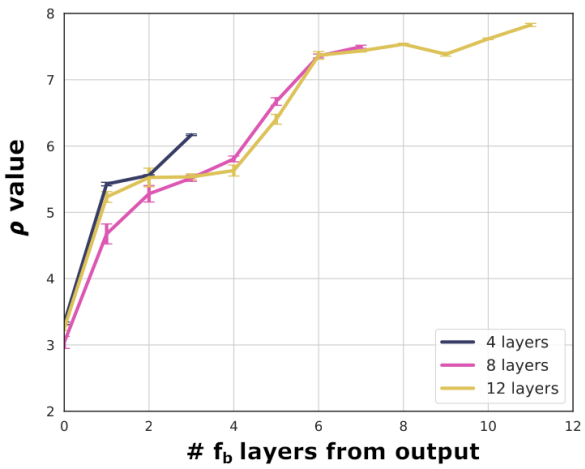

Fig. S2: **Values of ρ for varying sizes of $\widetilde{f_b}$ on ResNet-18 for CIFAR-10 data attacked with AutoAttack.** Observed ρ values tend to be slightly higher when f_b consists of 4 layers compared to 8 or 12 layers. The observed ρ values are highly similar when f_b consists of either 8 or 12 layers.

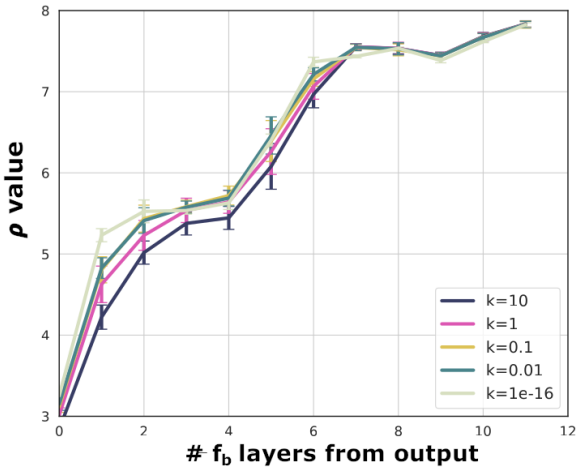

Fig. S3: **Values of ρ for varying k using ResNet-18 on CIFAR-10 data attacked with AutoAttack.** Observed ρ values tend to increase for smaller k values. This reflects stricter stopping conditions on the robustness of $\widetilde{f_b}$.

Applying Subnetwork Finetuning with Existing Methods: The theoretical framework laid out in this work describes the ability of robust subnetworks to help confer robustness to downstream layers. We demonstrate that this behavior can be observed or even improved when implementing existing methods for the adversarial training in Algorithm 1. We consider two methods to investigate their impact on Algorithm 1 and the values of ρ. ATTA [1] reduces the number of steps per PGD attack during training but allows perturbations to accumulate between epochs to improve efficiency, while TRADES [2] alters the loss function to balance adversarial and normal accuracy of the network. We implement ATTA with 3 steps per attack and reset perturbations every 10 epochs. For TRADES we use the implementation provided by [1], with standard PGD-10 perturbations and without resetting attacked data as is done for ATTA.

Table S5 shows these results for VGG16 on CIFAR-10 under

TABLE S5: **Algorithm 1 results using benchmark methods:** The summarized results of running Algorithm 1 for VGG16 on CIFAR-10 are shown when run with ATTA, TRADES, both, or standard adversarial training. Robust accuracy on test data attacked with AutoAttack was used to determine the values of Acc^* and $\widetilde{Acc}$. The difference in accuracy, the mean number of epochs of finetuning required to reach the baseline robust accuracy, and the mean trial runtime are recorded. The ρ values for a subset of the layers in $\widetilde{f}_b$ are reported, where ρ_n corresponds to the last pair of layers in $\widetilde{f}_b$.

Method	# f_b	Acc^*	$\widetilde{Acc}$	Diff.	Epochs	Time (sec)	ρ_n	ρ_{n-3}	ρ_{n-7}	ρ_{n-11}
PGD-10	4	0.62	0.1	-0.52	-	2347	2.71	6.94	-	-
	12	0.62	0.17	-0.45	-	2460	2.58	7.03	8.26	8.72
ATTA	4	4.66	2.95	-1.71	-	5510	2.84	6.78	-	-
	12	4.66	0.75	-3.91	-	4954	2.85	6.89	8.26	8.72
TRADES	4	29.2	29.74	0.53	2.4	1526	2.44	6.21	-	-
	12	29.2	25.2	-4.0	-	8754	2.42	5.76	7.33	8.02
ATTA+TRADES	4	30.67	30.3	-0.37	-	6531	2.64	6.35	-	-
	12	30.67	26.43	-4.24	-	8012	2.72	6.53	8.01	8.46

attack with AutoAttack using adversarial training with ATTA, TRADES, both, or standard adversarial training with PGD-10. We see that while ATTA requires fewer perturbations, it significantly under-performed compared to TRADES or using both together and as a result the overall runtime increases due to the larger number of required epochs. Notably the accuracy for PGD-10 is lower than observed in our own implementation of PGD-10 training seem in Table 1 of the main work. Ultimately these results demonstrate the impact that different methods may have on the ρ thresholds and ability of subnetwork f_a^* to confer robustness to f_b. We note that the lower performance of ATTA is likely due to the small number of epochs used per trial, which limit the ability for gradients to accumulate in the initial epochs.

We note that TRADES converged with the full network accuracy more rapidly in Table S5 for small subnetwork f_b sizes, whereas for standard adversarial training in Table 1 of the main work, we more frequently observe this for larger f_b. This potentially indicates that different loss functions may be more or less compatible with this transfer of robustness for a subnetwork of a given number of layers.

REFERENCES

[1] H. Zheng, Z. Zhang, J. Gu, H. Lee, and A. Prakash, "Efficient adversarial training with transferable adversarial examples," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 1181–1190.

[2] H. Zhang, Y. Yu, J. Jiao, E. Xing, L. El Ghaoui, and M. Jordan, "Theoretically principled trade-off between robustness and accuracy," in *International conference on machine learning*. PMLR, 2019, pp. 7472–7482.